

Deliverables

Student Name, Date, and Campus:

Bathsheba Myers May 15, 2026 Newport News ECPI

Place your numbered answers in this section. Refer to the activities above for complete instructions.

Deliverable #1: Standard PC Repair Toolkit

Hyperlink: [Amazon.com: Hi-Spec 56pc Electronics Repair & Opening Tool Kit Set for Laptops, Devices, Computers, PC Building & Gaming Accessories. Precision Small Screwdrivers with Pry Tools : Electronics](#)

Toolkit Name: Hi-Spec 56pc Electronics Repair & Opening Tool Kit Set

Price: \$37.99

Tools Included: Driver Handle Set, Precision Driver Bits, Full-Size Bits, Precision Screwdrivers, Pliers and Mini Wire Stripper, Pry Bar Levers and Suction Cup, Utility Knife, Tweezers & 3-Claw Pick Up Tool, Dust Brush, ESD Wrist Strap, Voltage Tester, Zip/Cable Ties, Magnetizing Block, Portable Zipper Case



Deliverable #2: ESD Safety

Definition of ESD: Electrostatic Discharge is ESD, which is the transfer of static electricity that buildup on an object that discharged to a different object resulting in damage to that another object.

Proper Safety Procedures: Be sure to use an ESD wrist strap, Ground workstations, handle the component by the edges, and use Ionizers if the environment is not possible to be grounded.

Purpose and Proper Use of Wrist Strap: The purpose of using a wrist Strap is designed to help safely discharge the electricity using the human body as a grounding point to prevent ESD damage to an object.

Deliverable #3: Screwdriver Types

Three standard screwdriver head types:

Phillips head Screwdriver



Flathead Screwdriver



Torx screwdriver



Non-standard/specialized screwdriver heads:

Precision Screwdrivers and Computer Screwdrivers



Deliverable #4: Mobile Device/Laptop Toolkit

Hyperlink: [Amazon.com: iFixit Pro Tech Toolkit - Electronics, Smartphone, Computer & Tablet Repair Kit : Electronics](https://www.amazon.com/dp/B078888888)

Toolkit Name: iFixit Pro Tech Toolkit

Price: \$79.95

Tools Included: Anti-Static Wrist strap, Small Suction cup, Opening Tool (x3), Nylon Tipped Reverse Tweezers, Angled Tweezers, Blunt ESD Tweezers, Spudger, Halberd Spudger, metal Spudger, Jimmy, Flex Extension, 64 Bit (4mm), Magnetic Driver, Tool Roll



Deliverable #5: Spudger Tool

What is a spudger: IT is a tool that has a flat wide head with a wedge type shape.

How it is used: A spudger is use for a mobile device that has a cracked screen that needs to be replaced, It may also be used for smartphone battery parts.

Spudger Tool Set



Image:

Deliverable #6: Ideal Toolkit Analysis

10 Essential Tools (with purpose/justification):

1. Magnetic Phillips Screwdriver - A screwdriver for tightening screws on your case, motherboard, and other components. A magnetic tip prevents screws from dropping into hard-to-reach spaces. A screwdriver is a must for any computer repair. Different computer models and components require screws of different types.
2. Precision Screwdriver – A must have for disassembling laptops, smartphones and other compact devices. Many electronic devices—smartphones, laptops, cameras, and tablets—use tiny screws that require precise tools to remove or tighten without damaging the components. Precision screwdrivers are designed with small, finely crafted tips to fit these tiny screws accurately.
3. Torx Drivers - (T5, T6, T8) are required for many laptops, SSD/HHD enclosures and Mac devices. Many electronic devices, including smartphones, tablets, laptops, gaming consoles, and automotive components, use Torx screws due to their unique star-shaped pattern. Standard Phillips or flat-head screwdrivers cannot fit these screws properly, increasing the risk of damage.
4. Anti-Static Wrist Strap (ESD) - Static electricity can permanently damage PC components. An anti-static wrist strap grounds you, preventing static discharge while handling sensitive parts. This is a must to safeguard the internal components of computers.
5. Compressed Air Canister - Dust can reduce component performance and airflow. Compressed air is perfect for cleaning your case, CPU cooler, and fans during or

after assembly. This is a must to help effectively keep the components clean and free from potential problems.

6. Cable Ties or Velcro Straps - Proper cable management is essential for both aesthetics and airflow within a computer case. Cable ties, including nylon zip ties and Velcro straps, help organize and secure cables neatly. Effective use of cable ties can significantly improve the internal appearance of a build and enhance airflow by reducing obstructions.
7. Needle nose pliers and wire cutters - Needle-nose pliers with their long and thin jaws, are essential for gripping and manipulating small components like jumpers or connectors. They allow for precise control in tight spaces. Wire cutters with built-in strippers are highly useful for cutting and stripping wires during repairs or customizations. These items are very useful for bend pin, trim zip ties, and cutting wires cleanly.
8. Plastic Spudgers – Essential for safely opening devices without scratching or cracking screens. These plastic or nylon tools are perfect for delicate operations like battery replacements.
9. Nylon or metal Tweezers - fine tips tweezers are perfect for handling tiny screws, pins, or delicate cables. They ensure a steady grip and prevent accidental damage to fragile components.
10. Allen Wrench set - An Allen wrench, or hex key, is used to tighten or loosen hexagonal socket screws that are often used in machinery assemblies and for fastening panels. This tool helps to secure parts in tight spaces, making accurate adjustments and assembling or disassembling components that use hex fasteners

5 Specialized/Supplemental Tools (with purpose/justification):

1. Multimeter – A multimeter is a versatile tool that allows you to measure voltage, current, and resistance. This tool is essential for troubleshooting power supply problems, faulty wiring, and other electrical components.
2. Cable Tester – Verify the integrity of Ethernet, USB and other data cables. This tool helps ensure that connectivity issues aren't caused by faulty wiring. Valuable tool for ensuring reliable and efficient network connections.

3. Magnetic Project Mat - Keep screws and components organized during disassembly. Having a mat helps to label sections, while reducing the risk of misplacement or confusion.
4. Thermal Paste - Thermal paste ensures efficient heat transfer between your CPU and cooler. This is crucial for ensuring optimal heat conduction between the CPU and its cooler because it improves cooling efficiency, leading to better performance and longevity of the processor.
5. Small flashlight - A small flashlight or headlamp is invaluable for illuminating dark corners inside a computer case. It helps with identifying and troubleshooting issues, especially in poorly lit areas. Illuminates dark areas inside the computer.

Selected Commercial Toolkit (hyperlink and description):

STREBITO Electronics Precision Screwdriver Sets 142-Piece with 120 Bits Magnetic Repair Tool Kit for iPhone, MacBook, Computer, Laptop, PC, Tablet, PS4, Xbox, Nintendo, Game Console

\$27.99

[Amazon.com: STREBITO Electronics Precision Screwdriver Sets 142-Piece with 120 Bits Magnetic Repair Tool Kit for iPhone, MacBook, Computer, Laptop, PC, Tablet, PS4, Xbox, Nintendo, Game Console : Electronics](#)



Deliverable #7: PC Hardware Familiarization

Online - PC Model Researched:

CLX SCARAB GAMING PC

Online - Key Specifications:

Component	Specification
Case Type	ATX Mid-Tower
CPU	AMD Ryzen 5 7600
CPU Cooler	DeepCool AK400 Air Cooler
Motherboard	MSI B650 ATX Motherboard
RAM	16GB DDR5 (2×8GB)
Storage	1TB NVMe SSD
Graphics Card	NVIDIA RTX 4060
Power Supply	750W 80+ Bronze PSU
Cooling	Front intake and rear exhaust fans

Operating System Windows 11

Online - Major Internal Components Expected:

Motherboard

- CPU socket
- RAM slots
- PCIe slots
- M.2 SSD slots
- SATA connectors
- Fan headers

2. CPU and Heatsink/Cooler

- AMD Ryzen 5 7600

3. RAM (Memory)

- 16GB DDR5 (2 sticks)

4. Graphics Card (GPU)

- NVIDIA RTX 4060

5. Storage Drives

- 1TB NVMe SSD

6. Power Supply Unit (PSU)

- 750W 80+ Bronze PSU

7. Cooling Fans

- CPU cooler fan
- Front intake fans
- Rear exhaust fan
- Optional top-mounted fans

8. Expansion Slots and Cards

- Graphics cards
- Wi-Fi adapters
- Sound cards
- Capture cards

9. Case Cables and Connectors

- 24-pin motherboard power cable
- CPU power cable
- GPU power cable
- SATA cables
- Front-panel connectors
- Fan cables

Online - Reference Images/Videos:

Best Custom Mini Gaming PC Builders | CLX Scarab



Deliverable #8: AI Supplement Findings

Tools listed from lab activity:

Comparison with project PC parts list:

AI-suggested additional tools:

Comparison and surprises:

Online toolkit search results and resources used: